Deep Reinforcement Learning

Notation

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Letter styles

- Sets are written in caligraphic upper-case, $\mathcal{A}, \mathcal{S}, \mathcal{O}, \ldots$.
- States, actions, or other elements of sets are written as lower-case letters, $s, a, o, \ldots$.
- The letter t is reserved for an iterator along a trajectories temporal dimension, its maximal value is given by T. Trajectories start at $t = 0$.
- We reserve i, j, k, n, m to label iterators, maximal values achieved by them are denoted by the corresponding upper-case letter, i.e., I, J, K, N, M.

MDPs

- An MDP is a tuple $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$. It is consists of a set of states $\mathcal{S}$ and actions $\mathcal{A}$. The transition distribution (discrete case) or density (continuous case) is denoted by $p(s'|s, a)$. Finally, we have a reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and a discount factor $\gamma \in [0, 1]$.
- For a given MDP, a policy $\pi(a|s)$ is either a distribution (discrete case) or a density (continuous case) over actions.
- Outputs are elements of some set $\mathcal{O}$, obtained by possibly implicit measurements of given state-action pairs, $o_t = h(s_t, a_t)$, where $h : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{O}$.

Random variables

- Random variables with distribution (discrete case) or density (continuous case) q are declared by writing $x \sim q$. E.g., $s' \sim p(\cdot|s, a)$ or $a \sim \pi(\cdot|s)$.
- Trajectories are sequences of state-action pairs

$$\tau = (\tau_0, \tau_1, \ldots) = ((s_0, a_0), (s_1, a_1), \ldots).$$

If the trajectory is sampled according to some policy π (i.e., $s_{t+1} \sim p(\cdot|s_t, a_t)$ and $a_t \sim \pi(\cdot|s_t)$ for $t \geq 1$), we get a new distribution/density, denoted by $p^\pi(\tau|s_0)$ which is conditioned on the initial state s_0. So $\tau \sim p^\pi(\cdot|s_0)$ denotes a random trajectory starting from s_0.

Expectation and variance

- Expectations are denoted as

$$\mathbb{E}_{x\sim q}[f(x)] = \underbrace{\mathbb{E}_x[f(x)]}_{\text{if distr. clear}} = \underbrace{\mathbb{E}[f(x)]}_{\text{if R.V. clear}} = \begin{cases} \sum_{x\in\mathcal{X}} q(x)f(x) & \text{discrete} \\ \int_{\mathcal{X}} q(x)f(x)dx & \text{continuous} \end{cases}$$

Similarly, the variance is defined by

$$\mathbb{E}_{x\sim q}[f(x)] = \mathbb{E}_x[f(x)] = \mathbb{E}[f(x)] = \mathbb{E}_x[(f(x) - \mathbb{E}_{x'}[f(x')])^2].$$

State value and state-action value

- The state-value function is defined as

$$V^\pi(s_0) := \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right], \quad \text{where} \quad s_{t+1} \sim p(\cdot|s_t, a_t), a_t \sim \pi(\cdot|s_t).$$

  alternatively, we can write

$$V^\pi(s_0) = \mathbb{E}_{\tau \sim p^\pi(\cdot|s_0)}\left[\sum_{t=0}^{\infty} \gamma^t r(\tau_t)\right].$$

- The state-action value function can be defined as

$$Q^\pi(s,a) = \mathbb{E}_{s' \sim p(\cdot|s,a)}[r(s,a) + \gamma V^\pi(s')].$$

Bellman equations

- We have the Bellman equations

$$\begin{aligned} V^\pi(s) &= \mathbb{E}_{a\sim\pi(\cdot|s),s'\sim p(\cdot|s,a)}[r(s,a)+\gamma V^\pi(s')] &&= \mathbb{E}_{a,s'}[r(s,a)+\gamma V^\pi(s')], \\ Q^\pi(s,a) &= \mathbb{E}_{s'\sim p(\cdot|s,a),a'\sim\pi(\cdot|s')}[r(s,a)+\gamma Q(s',a')] &&= \mathbb{E}_{s',a'}[r(s,a)+\gamma Q(s',a')]. \end{aligned}$$

- Optimal value functions are denoted by V^* or Q^*. They satisfy the Bellman optimality equations

$$\begin{aligned} V^*(s) &= \max_{a\in\mathcal{A}} \mathbb{E}_{s'}[r(s,a)+\gamma V^*(s')] \\ Q^*(s,a) &= \mathbb{E}_{s'}[r(s,a)+\gamma \max_{a'\in\mathcal{A}} Q^*(s',a')]. \end{aligned}$$

An optimal policy is denoted by π^*, satisfying $V^* = V^{\pi^*}$.

Trainable parameters and function approximation

- Trainable parameters are denoted by θ, ψ, ϕ. Dependence on θ for function approximators is denoted via subscript notation, i.e., V_θ^π, Q_θ^π, π_θ, etc. Differentiation w.r.t. a parameter is denoted by $\nabla_\theta(\cdot)$.
- Objective functions (for instance in the policy gradient algorithm) are denoted by J.
- Sequences of value function (approximations) are denoted by $V_0, V_1, V_2, \ldots$ when one has, for example, $V_n \to V^\pi$ as $n \to \infty$. (So iterator is in subscript notation.) Similarly, sequences of trainable parameters are denoted by $\theta_0, \theta_1, \theta_2, \ldots$.
- Datasets are denoted by $\mathcal{D}$, containing either states, state-action pairs, or other content gathered while sampling from an MDP.